

A Complemented Pair of Ideal-Null Spaces with Infinite-Dimensional Quotient

Candidate full solution packet

arXiv:2507.13866, Remark 6.7

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Status. Candidate full affirmative answer, subject to expert review. We construct proper ideals $\text{Fin} \subsetneq \mathcal{I} \subsetneq \mathcal{J}$ on $\mathbb{N}$ such that $c_{0,\mathcal{I}}$ is complemented in $c_{0,\mathcal{J}}$ and, more strongly,

$$c_{0,\mathcal{J}}/c_{0,\mathcal{I}} \text{ is linearly isometric to } \ell_\infty.$$

1 Source question

For an ideal $\mathcal{A}$ on $\mathbb{N}$, containing every finite set, put

$$c_{0,\mathcal{A}} = \{x \in \ell_\infty : \{k : |x(k)| \geq \varepsilon\} \in \mathcal{A} \text{ for every } \varepsilon > 0\}.$$

Rincón-Villamizar and Uzcátegui Aylwin study when $c_{0,\mathcal{I}}$ is complemented in $c_{0,\mathcal{J}}$ for proper ideals $\mathcal{I} \subsetneq \mathcal{J}$ [1]. After constructing examples with every finite quotient dimension, they ask in Remark 6.7 on printed page 17:

some infinite set $A \in \mathcal{J}$. Then, $\text{ad}_{\mathcal{I}}(\mathcal{J} \cap \mathcal{I}^+) > \aleph_0$.

An ideal $\mathcal{I}$ is *everywhere meager* [8], if $\mathcal{I} \restriction A$ is meager in 2^A for all $A \in \mathcal{I}^+$, which in turns is equivalent to requiring that $\mathcal{I} \restriction A$ is Baire measurable in 2^A for all $A \in \mathcal{I}^+$. Every analytic ideal is everywhere meager. On the opposite side, a complemented ideal $\mathcal{I}$ is nowhere meager, since $\mathcal{I} \restriction A$ is not meager in 2^A for all $A \in \mathcal{I}^+$ by Proposition 5.9.

From Theorem 6.3 we get the following corollary. A similar result appears in [20, Corollary 1.5] about the space c of convergent sequences.

Corollary 6.5. *Let $\mathcal{I} \subsetneq \mathcal{J}$ be proper ideals on $\mathbb{N}$ such that $\mathcal{I}$ is everywhere meager. Then $c_{0,\mathcal{I}}$ is not complemented in $c_{0,\mathcal{J}}$. In particular, c_0 is not complemented in $c_{0,\mathcal{J}}$ for any $\mathcal{J} \supsetneq \text{Fin}$.*

Proposition 6.6. *Let $\mathcal{I}$ and $\mathcal{J}$ be ideals on $\mathbb{N}$ with $\mathcal{I} \subsetneq \mathcal{J}$ and k be a positive integer. If $\mathcal{J}$ is complemented and $\mathcal{I}$ is k -maximal in $\mathcal{J}$, then $\mathcal{I}$ is complemented.*

Proof. Since $\mathcal{I}$ is k -maximal in $\mathcal{J}$, it follows that $c_{0,\mathcal{I}}$ is complemented in $c_{0,\mathcal{J}}$ by Theorem 4.6. Additionally, $c_{0,\mathcal{J}}$ is complemented in ℓ_∞ . Therefore, $c_{0,\mathcal{I}}$ is complemented in ℓ_∞ . This completes the proof. $\square$

Remark 6.7. Concerning the above results, Theorem 4.6 gives examples of proper ideals $\mathcal{I}$ and $\mathcal{J}$ with $\text{Fin} \subsetneq \mathcal{I} \subsetneq \mathcal{J}$ such that $c_{0,\mathcal{I}}$ is complemented in $c_{0,\mathcal{J}}$. Indeed, if $\mathcal{J}$ is a proper ideal and $\mathcal{I}_1, \dots, \mathcal{I}_k$ are maximal ideals such that $\mathcal{J} \not\subseteq \mathcal{I}_j$ for all $1 \leq j \leq k$, then $\mathcal{I} := \mathcal{J} \cap (\bigcap_{1 \leq j \leq k} \mathcal{I}_j)$ is k -maximal in $\mathcal{J}$. Thus, $c_{0,\mathcal{I}}$ is complemented in $c_{0,\mathcal{J}}$ by Theorem 4.6. Notice that in this case $\dim(c_{0,\mathcal{J}}/c_{0,\mathcal{I}}) = k$. We do not know if there exist examples with $\dim(c_{0,\mathcal{J}}/c_{0,\mathcal{I}}) = \infty$.

In spite of the previous results, it is still possible for $c_{0,\mathcal{I}}$ to contain complemented subspaces that are isomorphic or isometric to c_0 . In particular, we proved in [25] that $\ell_\infty(c_0)$ is isometric to c_{0,Fin^ω} , and since $\ell_\infty(c_0)$ clearly contains a complemented copy of c_0 , the same holds for c_{0,Fin^ω} . This contrasts with the situation in ℓ_∞ , where every subspace isomorphic or isometric to c_0 is necessarily not complemented. This leads to the following result.

Proposition 6.8. *Let $\mathcal{I}$ be an ideal on $\mathbb{N}$. If $c_{0,\mathcal{I}}$ has a complemented copy of c_0 , then $\mathcal{I}$ is not complemented.*

“We do not know if there exist examples with $\dim(c_{0,\mathcal{J}}/c_{0,\mathcal{I}}) = \infty$.”

2 Proof intuition

A maximal ideal is the complement of an ultrafilter, and its ideal-null space is the kernel of the corresponding ultralimit functional. Put countably many free ultrafilters on pairwise disjoint infinite blocks of $\mathbb{N}$. The map collecting their ultralimits is a contraction from ℓ_∞ to ℓ_∞ . It has an explicit isometric right inverse: make a sequence constant on each block.

Take $\mathcal{I}$ to be the intersection of all the associated maximal ideals and $\mathcal{J}$ to be one distinguished maximal ideal. On $c_{0,\mathcal{J}}$, the remaining ultralimits still map onto ℓ_∞ , their joint kernel is exactly $c_{0,\mathcal{I}}$, and the block-constant right inverse yields a projection of norm at most two. The same right inverse identifies the quotient norm exactly, so the quotient is not merely infinite-dimensional but isometric to ℓ_∞ .

3 Construction and proof

Partition $\mathbb{N}$ into pairwise disjoint infinite sets

$$\mathbb{N} = \bigsqcup_{n=0}^{\infty} A_n.$$

For each n , choose a free ultrafilter $\mathcal{V}_n$ on A_n and lift it to the ultrafilter $\mathcal{U}_n$ on $\mathbb{N}$ defined by

$$E \in \mathcal{U}_n \iff E \cap A_n \in \mathcal{V}_n.$$

Let $\mathcal{M}_n = \mathcal{P}(\mathbb{N}) \setminus \mathcal{U}_n$ be the corresponding maximal ideal, and define

$$\mathcal{J} = \mathcal{M}_0, \quad \mathcal{I} = \bigcap_{n=0}^{\infty} \mathcal{M}_n. \quad (1)$$

Lemma 1. *For $x \in \ell_\infty$, let*

$$\phi_n(x) = \lim_{\mathcal{U}_n} x.$$

Then ϕ_n is a norm-one linear functional and

$$c_{0,\mathcal{M}_n} = \ker \phi_n, \quad c_{0,\mathcal{I}} = \bigcap_{n=0}^{\infty} \ker \phi_n, \quad c_{0,\mathcal{J}} = \ker \phi_0. \quad (2)$$

Proof. The existence and linearity of the ultralimit of a bounded scalar sequence are standard, and $\phi_n(\mathbf{1}) = 1$ gives norm one. For $\varepsilon > 0$, the assertion $\phi_n(x) = 0$ is equivalent to

$$\{k : |x(k)| < \varepsilon\} \in \mathcal{U}_n,$$

for every $\varepsilon > 0$. Taking complements, this is equivalent to $\{k : |x(k)| \geq \varepsilon\} \in \mathcal{M}_n$ for every ε , which is precisely $x \in c_{0,\mathcal{M}_n}$. Intersecting these equivalences and using (1) proves the other two identities. $\square$

Theorem 2. *The ideals in (1) are proper and satisfy*

$$\text{Fin} \subsetneq \mathcal{I} \subsetneq \mathcal{J}.$$

There is a projection

$$P : c_{0,\mathcal{J}} \longrightarrow c_{0,\mathcal{I}} \quad \text{with} \quad \|P\| \leq 2,$$

and $c_{0,\mathcal{J}}/c_{0,\mathcal{I}}$ is linearly isometric to ℓ_∞ . Consequently the question in Remark 6.7 of [1] has an affirmative answer.

Proof. Each $\mathcal{M}_n$ contains Fin , since $\mathcal{U}_n$ is free. Choose one point $s_n \in A_n$ for each n and let $S = \{s_n : n \geq 0\}$. For every m , $S \cap A_m$ is a singleton and hence is not in the free ultrafilter $\mathcal{V}_m$. Thus $S \in \mathcal{M}_m$ for all m , so the infinite set S belongs to $\mathcal{I}$ and $\text{Fin} \subsetneq \mathcal{I}$. Moreover, $A_1 \in \mathcal{M}_0 = \mathcal{J}$ but $A_1 \notin \mathcal{M}_1$, giving $\mathcal{I} \subsetneq \mathcal{J}$. Both ideals are proper because $\mathbb{N}$ belongs to every $\mathcal{U}_n$.

Define

$$L : c_{0,\mathcal{I}} = \ker \phi_0 \longrightarrow \ell_\infty(\{1, 2, \dots\}), \quad Lx = (\phi_n(x))_{n \geq 1}. \quad (3)$$

This is a contraction. For $y = (y_n)_{n \geq 1} \in \ell_\infty$, define the block-constant sequence $Ry \in \ell_\infty(\mathbb{N})$ by

$$(Ry)(k) = \begin{cases} 0, & k \in A_0, \\ y_n, & k \in A_n, \ n \geq 1. \end{cases} \quad (4)$$

Then $\|Ry\| = \|y\|$, $\phi_0(Ry) = 0$, and $\phi_n(Ry) = y_n$ for $n \geq 1$. Therefore R is an isometric right inverse of L whose range lies in $c_{0,\mathcal{I}}$.

By Lemma 1,

$$\ker L = \bigcap_{n=0}^{\infty} \ker \phi_n = c_{0,\mathcal{I}}.$$

It follows that

$$P = I_{c_{0,\mathcal{I}}} - RL \quad (5)$$

is a projection onto $c_{0,\mathcal{I}}$ and $\|P\| \leq 2$.

Finally, let $\tilde{L} : c_{0,\mathcal{I}}/c_{0,\mathcal{I}} \rightarrow \ell_\infty$ be the map induced by L . If $x \in c_{0,\mathcal{I}}$ and $z \in c_{0,\mathcal{I}}$, then

$$\|Lx\| = \|L(x+z)\| \leq \|x+z\|,$$

so $\|Lx\| \leq \|x + c_{0,\mathcal{I}}\|$. Conversely, $x - RLx = Px \in c_{0,\mathcal{I}}$, and hence

$$\|x + c_{0,\mathcal{I}}\| \leq \|RLx\| = \|Lx\|.$$

Thus $\tilde{L}$ is an onto linear isometry, completing the proof. $\square$

4 Verification and limitations

The construction uses free ultrafilters, hence the usual ultrafilter lemma available in ZFC. It gives an explicit relative projection of norm at most two and identifies the quotient isometrically; no separability, positivity of the relative projection, or definability of the ideals is claimed. Indeed the ideals are necessarily nonmeager and highly nondefinable, in line with the obstructions developed in the source.

The main verifier checks are: (i) the level-set definition of $c_{0,\mathcal{M}_n}$ agrees with the kernel of the scalar ultralimit; (ii) the block-constant right inverse lies in $\ker \phi_0$; and (iii) the two quotient norm inequalities have the correct directions. All three are explicit in the proof.

A bounded novelty search through 11 August 2026 checked the run's registry, solution, attempt, and proof-gap indexes; arXiv:2507.13866v2; exact-title, exact-question, author, ideal-null, relative-complementation, and infinite-quotient searches. The current v2, revised 18 March 2026, still contains the question. No later paper or independent answer was located. Novelty confidence is moderate, because bounded search is not a proof of priority.

5 Human-review recommendation

Recommended for expert review as a full solution. The proof is short and self-contained; review should focus on whether the authors intended any unstated regularity restriction on the ideals. Under their printed definition of proper ideals containing Fin , the construction meets the question exactly.

References

- [1] M. A. Rincón-Villamizar and C. Uzcátegui Aylwin, *On the complementation of spaces of $\mathcal{I}$ -null sequences*, arXiv:2507.13866v2, 2026; Remark 6.7, p. 17.